

Positive Tutte Evaluations and an Explicit Finite Count

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October 5, 2026

Abstract

For every finite undirected multigraph G and integer $q \geq 2$, we prove $T_G(q+1, q-1) \geq q^{r(E)} > 0$. We construct a finite family of decorated spanning subgraphs whose cardinality is this evaluation and give a terminating algorithm to enumerate it. This proves the positivity and effective-count assertions in conjecture 00000000585, with no prime-power restriction needed.

1 Statement and conventions

The published conjecture states: “For every prime power q , $T_G(q+1, q-1) > 0$ and yields an effective count (an arithmetization of the combinatorial positivity interpretation).” Its preceding descriptive sentence mentions log-concavity but specifies no sequence, inequality, or additional log-concavity assertion. The theorem below addresses exactly the stated positivity and counting assertions; it does not assert an unspecified log-concavity property.

Let $G = (V, E)$ be a finite undirected graph. Loops, parallel edges, isolated vertices, disconnected graphs, and the graph with no vertices are permitted. Edges are individually labeled; consequently parallel edges determine different subsets of E . For $A \subseteq E$, write $c(A)$ for the number of connected components of the spanning graph (V, A) , including isolated vertices, and put

$$r(A) = |V| - c(A), \quad \nu(A) = |A| - r(A).$$

For the empty vertex set the component count is zero. In particular $r(\emptyset) = 0$. We use the standard rank-subset definition of the Tutte polynomial:

$$T_G(x, y) = \sum_{A \subseteq E} (x-1)^{r(E)-r(A)} (y-1)^{|A|-r(A)}. \tag{1}$$

All exponents in this polynomial are nonnegative integers. Indeed, adding edges cannot increase the number of components, so $r(A) \leq r(E)$. Starting with $|V|$ isolated vertices, each inserted edge decreases the component count by at most one, so $r(A) \leq |A|$. These observations cover loops and parallel edges without any change. As usual a zeroth power is one, including $0^0 = 1$.

2 Positivity

Theorem 1. *For every finite undirected multigraph G and every integer $q \geq 2$,*

$$T_G(q+1, q-1) = \sum_{A \subseteq E} q^{r(E)-r(A)}(q-2)^{\nu(A)} \geq q^{r(E)} > 0.$$

Proof. Substitute $x = q+1$ and $y = q-1$ into (1). Since $q \geq 2$, the bases q and $q-2$ are nonnegative. Every summand is therefore nonnegative. For $A = \emptyset$, the summand is $q^{r(E)-0}(q-2)^0 = q^{r(E)}$, which is strictly positive. Keeping that summand and discarding the other nonnegative summands proves the assertion. $\square$

If $q = p^k$ is a prime power, with p a prime and $k \geq 1$, then $q \geq 2$. The conjectured positivity is thus an immediate instance of Theorem 1. The stronger conclusion also includes integers that are not prime powers.

3 An explicit counting interpretation

For an integer $s \geq 0$, let $[s] = \{0, \dots, s-1\}$. The set of words of length d over this alphabet is $[s]^d$, the set of functions from $[d]$ to $[s]$. Its cardinality is s^d . In particular there is one empty word over an empty alphabet and no positive-length word over that alphabet.

Define the finite set

$$\mathcal{C}(G, q) = \coprod_{A \subseteq E} (\{A\} \times [q]^{r(E)-r(A)} \times [q-2]^{\nu(A)}). \quad (2)$$

An object is a selected spanning subgraph, a q -color word in its rank-deficiency slots, and a $(q-2)$ -color word in its cycle-nullity slots. The selected edge set is part of the object, so contributions from different subsets cannot be confused. No equivalence of colorings is taken.

Theorem 2. *For every G and every integer $q \geq 2$,*

$$|\mathcal{C}(G, q)| = T_G(q+1, q-1).$$

The family $\mathcal{C}(G, q)$ is nonempty and has a finite, explicit enumeration algorithm.

Proof. The summand indexed by A in (2) has cardinality $q^{r(E)-r(A)}(q-2)^{\nu(A)}$, by the product rule for cardinalities. Taking the disjoint union gives the sum in Theorem 1. The empty edge set supplies $q^{r(E)}$ objects, so the family is nonempty.

To enumerate it, order the vertices and individually labeled edges. Compute $c(E)$ by graph search. Enumerate the $2^{|E|}$ Boolean edge-selection masks. For each mask, compute $c(A)$ by graph search, calculate the two displayed nonnegative lengths, and enumerate the words over the corresponding finite alphabets in lexicographic order. Output the mask and the two words. Every loop of this procedure has a specified finite bound. Each valid object appears exactly once, because its mask and each word have unique encodings. Thus this is a terminating effective enumeration. Counting instead of materializing the output gives the same evaluation. No polynomial-time claim is made. $\square$

At $q = 2$, precisely the acyclic subsets contribute: if $\nu(A) > 0$, their contribution is zero; if $\nu(A) = 0$, it is $2^{r(E)-r(A)}$. The empty subset still contributes positively. This also explains why the endpoint $q = 2$ does not present an exception.

4 Lean definitions and scope

The accompanying `Tutte585.lean` uses Lean 4.31.0 and its standard library only. The complete Lake project and executable examples compile successfully. The main theorem’s audited dependencies are `propext`, `Classical.choice`, and `Quot.sound`; no additional axiom or proof placeholder occurs. A graph has a finite vertex type $\text{Fin}(n)$ and a list of pairs of vertices. List positions are edge labels, so repeated endpoint pairs represent parallel edges. Equal endpoints represent loops.

The inductive relation `Connected` is generated by reflexivity, edges, symmetry, and transitivity. The definition `componentCount` counts the least vertex in each connected component. Each nonempty component has exactly one such least vertex, so this is precisely $c(A)$. The definition `rank` is $n - \text{componentCount}$, and `tutte` is exactly the rank-subset expansion above, evaluated in the natural numbers for $x, y \geq 1$. Only $x = q + 1$, $y = q - 1$ with $q \geq 2$ is claimed to represent the usual polynomial evaluation. Natural subtraction in the exponents equals integer subtraction by the graph inequalities proved in Section 1; these basic rank inequalities are not separately formalized in this file.

The formal proof establishes empty-graph connectivity as equality and $r(\emptyset) = 0$, proves the lower bound and strict positivity, and proves that the explicitly defined decorated-subgraph enumeration has the same length as the evaluation and has no duplicate objects. Decorations are represented by pairs of integers less than the corresponding powers; these are the ordinary finite word codes. Boolean masks preserve edge identity throughout enumeration. The main theorem is `conjecture_00000000585`.

The formal component count is executable. A finite cut test enumerates all vertex subsets and asks whether every edge-closed subset containing u also contains v . The file proves that this test is equivalent to `Connected`: connectivity preserves membership in a closed cut; conversely, the component of u itself is an edge-closed cut containing u . Classical reasoning is used only inside this correctness proof. The decision procedure, component count, rank, and object enumeration are all computable Lean definitions. This exhaustive cut procedure is deliberately simple and may be exponential in the number of vertices; graph search gives the more practical implementation described above.

There are no added axioms or proof placeholders. The final source prints its theorem’s logical dependencies for audit. In particular, ordinary classical logical dependencies are not graph-theoretic assumptions.

Source

The Last Math Competition, conjecture 00000000585, revision `d7d4bc9f6918`. Published statement at the exact pinned revision.